

Terraform Assignment: Modular Infrastructure for Azure Web Apps

Scenario

You are an Infrastructure Engineer tasked with provisioning cloud infrastructure and deploying a simple web application across **three isolated environments**: dev, uat, and prod.

Your goal is to design a **modular Terraform solution** that allows easy provisioning, maintenance, and scaling of each environment independently.

Assignment Objectives

1. Provision infrastructure in **Azure** using **Terraform modules**.
2. Deploy a simple **Apache-based web application** using **Azure Web Apps**.
3. Ensure each environment is **fully isolated** and independently accessible.
4. Structure your Terraform code to support **reusability and maintainability**.

Requirements

Infrastructure (per environment)

Each environment must include:

- A **Resource Group**
- An **App Service Plan**
- An **Azure Web App** running an official Apache HTTP Server container (e.g., httpd from Docker Hub)
- A **configuration setting** or environment variable that sets the index page to display: *"This is <env-name> environment."*

Each Web App must be publicly accessible via:

- `http://<dev-app>.azurewebsites.net`
- `http://<uat-app>.azurewebsites.net`
- `http://<prod-app>.azurewebsites.net`

Modularity Requirements

Your Terraform code **must** use modules to encapsulate reusable logic. Suggested structure:

```
terraform/
├─ main.tf
├─ variables.tf
├─ outputs.tf
├─ environments/
│   ├─ dev/
│   │   └─ main.tf
│   ├─ uat/
│   │   └─ main.tf
│   └─ prod/
│       └─ main.tf
└─ modules/
    └─ webapp/
        ├─ main.tf
        ├─ variables.tf
        └─ outputs.tf
```

Suggested Module Scope

modules/webapp/ should encapsulate:

- Resource Group creation
- App Service Plan
- Web App deployment
- Configuration settings (e.g., environment message)

Each environment (dev, uat, prod) should call the module with different parameters:

- Environment name
- Resource naming prefix
- Location
- App settings (e.g., message)

Deliverables

- Terraform code with modular structure
- Sample output of terraform plan and terraform apply
- Screenshot or curl output showing correct message from each Web App
- A **README** explaining:
 - How to deploy each environment
 - How modules are structured
 - Any assumptions or decisions made